Charbel Fayad

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Experience

Software Integration Engineer, Murex 06/2024 – Present

- Architect API-led and event-driven integrations on MuleSoft Anypoint Platform that connect 20+ cloud and on-premise systems, exchanging real-time and asynchronous messages through ActiveMQ and RabbitMQ.
- Apply enterprise integration patterns such as canonical data models, content-based routers and API gateways, and design for high availability with clustering and active-passive failover.
- Built from scratch an Anypoint management-as-code tool that keeps API contracts and specifications, API instances and gateways under version control and deploys them automatically through the CI/CD pipeline.
- Built an enterprise observability tool for ActiveMQ in React, PostgreSQL and Docker, giving the team queue management with end-to-end message traceability and audit trails.
- Design business-process-driven REST and SOAP APIs from stakeholder requirements, and keep them reliable in production through structured logging with Log4j and proactive monitoring in Splunk.
- Maintain Jenkins pipelines in Groovy for Maven and Dockerized Node.js projects, with artifacts on Nexus and Harbor, dependency scanning with Black Duck and container image scanning with Prisma Cloud.
- *Scrum Master:* facilitate Scrum ceremonies within SAFe, coordinate cross-team dependencies, mitigate delivery risks and present team deliveries.
- *Support Coordinator:* triage support requests, analyze recurring issues and coordinate fixes across the team.

Projects

SD-WAN and SASE Orchestrator 2026

- Architected B2B2B multi-tenant orchestration on a custom Linux NFV infrastructure over KVM, with near-zero-touch edge provisioning and service-chained VNFs for SD-WAN, firewalling and inline IDS/IPS.
- Architected the telemetry services that stream data from thousands of IoT edge devices through Kafka into Apache Flink for real-time processing and ClickHouse for analytics, replicated across availability zones for fault tolerance.
- Built edge agents in Go that report to regional control planes over gRPC, running on AWS and Kubernetes, with the data plane deployed at the edge or in the cloud.

Bishara, iOS 2026

- Arabic scripture on the Lock Screen, widgets and wallpapers. Built solo in React Native and Expo with native widget extensions; fully offline, no accounts.

MulePrep, **Petale**, **Rukhsa** 2026

- MuleSoft certification practice engine in React on Cloudflare's edge; a bouquet composer in React and Vite that serializes arrangements into shareable links; a multilingual PWA for Lebanese driving-theory exams.

Socrat 2024

- Open-source WebRTC platform on Next.js for virtual Socratic dialogue, with real-time participation and attendance analytics.

Earlier work 2022 – 2024

- Tourify, a tourism app in Ionic, Angular, PHP and MySQL; a dollar-rate API in FastAPI and a Flutter/Dart client; Smoll on SolidJS and MongoDB; a Mandelbrot renderer in C#; a terminal-style portfolio in Next.js, TypeScript and Tailwind CSS.

Research

Machine Learning Research, Lebanese American University 03/2022 – 06/2025

- Designed a genetic algorithm whose population evolves on a dynamic scale-free (Barabási–Albert) network, where authority nodes balance exploration and exploitation, and a neural network ensemble embedded in a scale-free graph, weighted by node centrality and pruned through its power-law degree distribution.
- Developed adaptive-topology neural networks that rewire low-importance connections during training, and a genetic programming approach to federated learning on binary neural networks, in Python with TensorFlow, Keras, DEAP and NetworkX.
- Fayad, C., & Eid, C. *A Scale-Free Network for Modeling a Homogeneous Ensemble*. Authorea preprint, 2026.
- Fayad, C. *Adaptive Neural Topologies: A Dynamic Approach to Learning Network Structures during Training*. TechRxiv preprint, 2025.
- Hoblos, Y., & Fayad, C. *A Scale-Free Network-Based Genetic Algorithm with Balanced Exploration and Exploitation*. engrXiv preprint, 2025.

Education

Lebanese American University, Byblos 09/2021 – 06/2024

BS Computer Science, Honors Program. CGPA 3.97, Major GPA 4.0, High Distinction.

Extracurricular

- Founder, Software Engineering Club, LAU: workshops and technical seminars.
- Founder, Computer Science Club, LAU.
- Lab Instructor, LAU: Operating Systems and Computer Programming labs.